1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 analysis identifies an apparent paradox — imaging/pathology–inferred immune infiltration (in-situ TILs) should predict better survival after PARP inhibition, yet some PARP-treated, BRCA1/2–deficient tumors do not derive OS benefit. The Ontologist's core paths link PARP therapy → immune contexture (pathomics/radiomics) and BRCA1/2 → PD-L1 → T-cell exhaustion → PTGER4, suggesting an immunosuppressive prostaglandin axis might uncouple immune presence from immune efficacy?

Working Hypothesis

In BRCA1/2-mutant metastatic castration-resistant prostate cancer (mCRPC) treated with Olaparib, high intratumoral TIL density is associated with improved overall survival (OS) only when tumor PTGER4 expression is low. Elevated PTGER4 expression attenuates the OS benefit of TIL-rich tumors by promoting PGE2–PTGER4–driven T-cell exhaustion. Consequently, in BRCA1/2-mutant mCRPC, the prognostic value of TIL density for OS is modulated by PTGER4 levels, such that: (a) TIL-high/PTGER4-low tumors show the strongest OS benefit from Olaparib; (b) TIL-high/PTGER4-high tumors derive reduced OS benefit, potentially nullifying the advantage conferred by TIL density alone. This mechanism predicts that adding PTGER4 blockade to PARP inhibition will restore or enhance the OS benefit in TIL-high, PTGER4-high tumors, effectively converting non-responders to responders within this molecular context. The hypothesis therefore supports a biomarker-guided combination strategy (PARP inhibition plus PTGER4 antagonism) in select BRCA1/2-mutant patients to improve survival outcomes.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PTGER4, BRCA1/2
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PTGER4: PTGER4, suggesting an immunosuppressive prostaglandin axis might uncouple immune presence from immune efficacy.

BRCA1/2: BRCA1/2 → PD-L1 → T-cell exhaustion → PTGER4, suggesting an immunosuppressive prostaglandin axis might uncouple immune presence from immune efficacy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PD-L1 (CD274)	Supporting modifier from knowledge graph

Secondary	T-cell exhaustion markers (PDCD1, LAG3)	Supporting modifier from knowledge graph
Secondary	CENPA	Supporting modifier from knowledge graph
Secondary	urinary PCA3	Supporting modifier from knowledge graph
Secondary	PSA nadir <0.2 ng/mL at 6 months	Supporting modifier from knowledge graph
Secondary	EDIH / systemic inflammatory biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: BRCA1/2 mutations confer sensitivity to PARP inhibitors but OS benefits are variable, suggesting contextual factors modulate efficacy.

Prediction 2: TIL density correlates with prognosis in several cancers and can be imaged or histologically quantified, yet immune suppression–mediated exhaustion can negate beneficial effects of infiltration.

Prediction 3: PTGER4, the receptor for PGE2, is a key mediator of prostaglandin-driven immunosuppression and T-cell exhaustion; high PTGER4 signaling can impair effective anti-tumor immunity even in TIL-rich tumors.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PD-L1 (CD274): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- T-cell exhaustion markers (PDCD1, LAG3): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.